

# Giorgos Kritikakis

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## SUMMARY

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Giorgos (George) Kritikakis is a full-stack software engineer at [Tom Sawyer Software](#), where his work includes designing and developing visual query builders, advanced data federation and ETL tools, and AI-powered features that enable users to integrate and explore complex data without needing to understand the underlying technologies. He also has a strong research background focused on developing practical solutions to challenging computational problems, with work published in leading international research venues. He holds a BSc and an MSc in Computer Science.

## PROFESSIONAL EXPERIENCE

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### Product Developer

#### Tom Sawyer Software

Mar 2023 - Present

Designs, develops, and tests product features across data integration and querying workflows. Key contributions include developing visual query builders and query-generation tools for graph databases, implementing AI-powered text-to-query features, adding support for MySQL, Microsoft SQL Server, IBM Db2, and Oracle Database, contributing to TS Data Streams, and developing internal benchmarking and profiling tools.

Selected demos: [Advanced Query Builder](#) | [Textual Query Builder](#) | [Natural Language Query](#) | [Explorations](#)

### Associate researcher

#### University of Crete - Graph and Information Visualization Laboratory

2020 - 2023

In this role, he designed and developed advanced graph algorithms for path/chain decomposition, transitive closure and reachability, and hierarchical graph visualization. The work has applications in areas such as software and task dependency analysis, citation and web networks, knowledge graphs, biological networks, and resource scheduling. He introduced new algorithms and indexing techniques, evaluated them extensively through experiments, and published the results in peer-reviewed venues.

### IT support

#### Greek Army - SEAP Heraklion

Mar 2022 - Dec 2022

Maintain all the systems, the servers, and the networks in the camp.

### Teaching Assistant

#### University of Crete

2020 - 2022

- CS-484 Complex Network Dynamics [Spring 2021]
- CS-380 Algorithms and Complexity [Fall 2021, Fall 2020]
- CS-486 Principles of Distributed Computing [Spring 2020]

### Associate Researcher

#### Institute of Computer Science (ICS), CARV Laboratory

2019 - 2020

Extension of the SCOOP compiler source-to-source C code transformations. SCOOP produces output for the PARTEE runtime system, a project developed at the Computer Architecture and VLSI Systems (CARV) Laboratory of the Institute of Computer Science (ICS) of the Foundation of Research and Technology Hellas (FORTH). PARTEE runtime uses annotations to specify tasks and their memory footprints. Scoop enables us to use pragma directives for the task annotation.

## EDUCATION

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2022 M.Sc. (Computer Science) at **University of Crete**

Thesis: Analysis and Visualization of Hierarchical Graphs.

Area of Study: *a)* Algorithms and Systems Analysis, *b)* Parallel and Distributed Systems.

2020 B.Sc. (Computer Science) at **University of Crete**

Thesis: Extension of the PARTEE runtime system with support for dynamic memory allocation.

Area of Study: Software Systems and Applications..

## TECHNICAL SKILLS

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- **Programming & Software Development:** Java, C++, C, Python, JavaScript, TypeScript, OCaml, SQL, HTML, HTML5, CSS; Object-Oriented Programming (OOP), Software Design Patterns, Systems Design, Distributed Systems, Multithreading, Parallel Programming, Compilers, Algorithms, Data Structures; Software Development, Problem Solving
- **Frameworks, Libraries & APIs:** Spring Boot, Spring Framework, Hibernate, JPA, React.js; REST, LLM Integration, ChatGPT, Prompt Engineering & Response Parsing
- **Databases & Data Skills:** Neo4j, Gremlin Query Language, Cypher Query Language, SQL; Data Analysis, Data Visualization
- **Testing, Tools & DevOps:** JUnit, Mockito, Selenium, Regression Testing; Git, Bitbucket, Jira, Maven, Docker Products, Shell Scripting, Windows, Linux
- **Professional & Interpersonal Skills:** Agile Methodologies, Project Management, Product Road Mapping, Communication, Intercultural Skills, Teaching Assistant, Research Skills

## PUBLICATIONS

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- Parameterized Linear Time Transitive Closure. Giorgos Kritikakis, and Ioannis G. Tollis. Journal of Graph Algorithms and Applications (JGAA).
- Fast Reachability Using DAG Decomposition. Giorgos Kritikakis, and Ioannis G. Tollis. 21st International Symposium on Experimental Algorithms (SEA 2023).
- Experiments and a User Study for Hierarchical Drawings of Graphs. Panagiotis Lionakis, Giorgos Kritikakis, and Ioannis G. Tollis. IEEE Access. May 29, 2023
- Fast and Practical DAG Decomposition with Reachability Applications. Giorgos Kritikakis, and Ioannis G. Tollis (ArXiv Dec 2022).
- Experiments and a User Study for Hierarchical Drawings of Graphs. Panagiotis Lionakis, Giorgos Kritikakis, and Ioannis G. Tollis (ArXiv Sep 2022).
- Algorithms and Experiments using the Path-Based Hierarchical Drawing Framework. Panagiotis Lionakis, Giorgos Kritikakis, and Ioannis G. Tollis.  
This work has been presented as a poster at 28th International Symposium on Graph Drawing and Network Visualization, GD 2020).

## ADDITIONAL INFO

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He has completed more than 200 hours of training through online platforms and seminars on topics including Generative AI, RAG, LoRA and AI Agents, Software Architecture, System Design & Big Data Processing, Spring Boot, IoT, wireless communication, Arduino, Raspberry Pi, ESP boards, version control systems (Git/GitHub), Linux inter-process communication, and web development.

In addition to his technical background, he has been a certified professional lifeguard since 2015. He also earned the SSI Open Water Diver certification in October 2023.